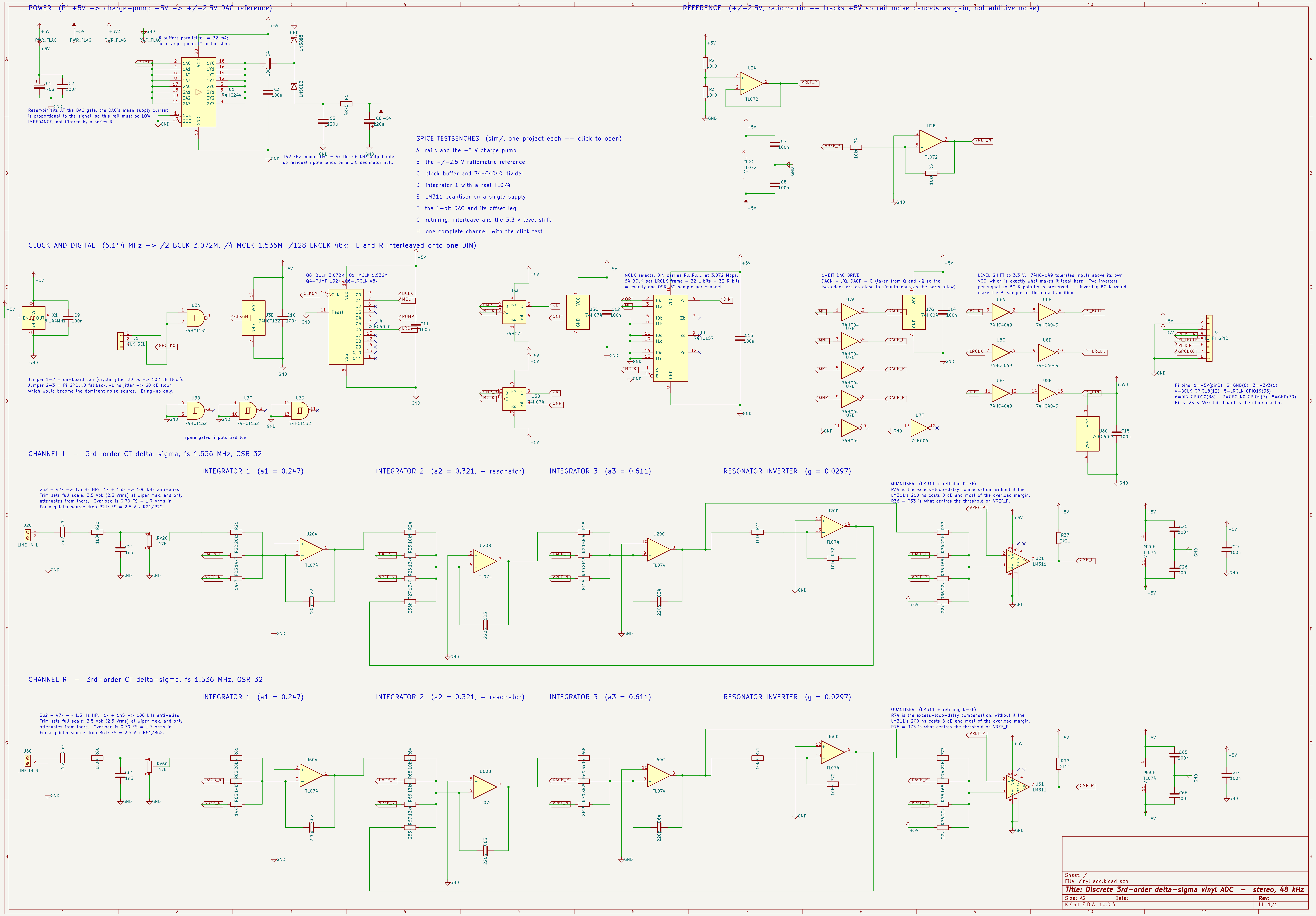
POWER (Pi +5V -> charge-pump -5V -> +/-2.5V DAC reference)
8 buffers paralleled ~= 32 mA;
no charge-pump IC in the shop
Reservoir sits at the DAC gate: the DAC's mean supply current
is proportional to the signal, so this rail must be LOW
IMPEDANCE, not filtered by a series R.
192 kHz pump drive = 4x the 48 kHz output rate,
so residual ripple lands on a CIC decimator null.
REFERENCE (+/-2.5V, ratiometric -- tracks +5V so rail noise cancels as gain, not additive noise)
SPICE TESTBENCHES (sim/, one project each -- click to open)
A rails and the -5 V charge pump
B the +/-2.5 V ratiometric reference
C clock buffer and 74HC4040 divider
D integrator 1 with a real TL074
E LM311 quantiser on a single supply
F the 1-bit DAC and its offset leg
G retiming, interleave and the 3.3 V level shift
H one complete channel, with the click test
CLOCK AND DIGITAL (6.144 MHz -> /2 BCLK 3.072M, /4 MCLK 1.536M, /128 LRCLK 48k; L and R interleaved onto one DIN)
Q0=BCLK 3.072M Q1=MCLK 1.536M
Q4=PUMP 192k Q6=LRCLK 48k
Jumper 1-2 = on-board can (crystal jitter 20 ps -> 102 dB floor).
Jumper 2-3 = Pi GPCLK0 fallback: ~1 ns jitter -> 68 dB floor,
which would become the dominant noise source. Bring-up only.
spare gates: inputs tied low
MCLK selects: DIN carries R,L,R,L... at 3.072 Mbps.
64 BCLK per LRCLK frame = 32 L bits + 32 R bits
= exactly one OSR=32 sample per channel.
1-BIT DAC DRIVE
DACN = /Q, DACP = Q (taken from Q and /Q so the
two edges are as close to simultaneous as the parts allow)
LEVEL SHIFT to 3.3 V. 74HC4049 tolerates inputs above its own
VCC, which is exactly what makes it legal here. Two inverters
per signal so BCLK polarity is preserved -- inverting BCLK would
make the Pi sample on the data transition.
Pi pins: 1=+5V(pin2) 2=GND(6) 3=+3V3(1)
4=BCLK GPIO18(12) 5=LRCLK GPIO19(35)
6=DIN GPIO20(38) 7=GPCLK0 GPIO4(7) 8=GND(39)
Pi is I2S SLAVE: this board is the clock master.
CHANNEL L - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32
INTEGRATOR 1 (a1 = 0.247)
INTEGRATOR 2 (a2 = 0.321, + resonator)
INTEGRATOR 3 (a3 = 0.611)
RESONATOR INVERTER (g = 0.0297)
2u2 + 47k -> 1.5 Hz HP: 1k + 1n5 -> 106 kHz anti-alias.
Trim sets full scale: 3.5 Vpk (2.5 Vrms) at wiper max, and only
attenuates from there. Overload is 0.70 FS = 1.7 Vrms in.
For a quieter source drop R21: FS = 2.5 V x R21/R22.
QUANTISER (LM311 + retiming D-FF)
R34 is the excess-loop-delay compensation: without it the
LM311's 200 ns costs 8 dB and most of the overload margin.
R36 = R33 is what centres the threshold on VREF_P.
CHANNEL R - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32
INTEGRATOR 1 (a1 = 0.247)
INTEGRATOR 2 (a2 = 0.321, + resonator)
INTEGRATOR 3 (a3 = 0.611)
RESONATOR INVERTER (g = 0.0297)
2u2 + 47k -> 1.5 Hz HP: 1k + 1n5 -> 106 kHz anti-alias.
Trim sets full scale: 3.5 Vpk (2.5 Vrms) at wiper max, and only
attenuates from there. Overload is 0.70 FS = 1.7 Vrms in.
For a quieter source drop R61: FS = 2.5 V x R61/R62.
QUANTISER (LM311 + retiming D-FF)
R74 is the excess-loop-delay compensation: without it the
LM311's 200 ns costs 8 dB and most of the overload margin.
R76 = R73 is what centres the threshold on VREF_P.
Sheet: /
File: vinyl_adc.kicad_sch
Title: Discrete 3rd-order delta-sigma vinyl ADC - stereo, 48 kHz
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1